

Tarun Mittal

Senior Software Engineer

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SUMMARY

- Senior Software Engineer with **5 years of experience** designing and owning cloud-native distributed systems.
- Expertise in **Python microservices**, OLAP data platforms (**ClickHouse**, **PostgreSQL**), event-driven systems, and **Agentic AI and RAG**.
- Led backend migrations end to end and **mentored engineers** at Uber, Masters India, and Impact Analytics.

SKILLS

- Languages:** Python, Go (Golang), SQL.
- Backend & APIs:** FastAPI, Gin, Django, SQLAlchemy, Pydantic, REST, gRPC, Celery, asyncio.
- AI & Agents:** LLM agents (OpenAI/Gemini/Claude), LangGraph, LangChain, MCP, RAG, pgvector.
- Data & Streaming:** Kafka, Flink, Spark, Pinot, ClickHouse, BigQuery, PostgreSQL, MySQL, Redis, MongoDB, Elasticsearch, S3.
- Cloud & Infra:** GCP, AWS, Docker, Kubernetes, CI/CD, Bazel, Observability (ELK, New Relic, Prometheus, Grafana).
- Core:** Distributed Systems, Microservices, System Design (HLD/LLD), Concurrency, High Availability, Caching, Mentorship, Unit Testing.

PROFESSIONAL EXPERIENCE

Senior Software Engineer IMPACT ANALYTICS, Bangalore May 2026 – Present

Agentic AssortSmart (AI Powered Retail Merchandise Planning Platform)

- Building the agentic rebuild of AssortSmart with a **LangGraph and MCP copilot** that plans store clusters through **14 audited tools**, targeting plan turnaround **under 1 hour** from days and failures under 2% from 8.5%.
- Led the **PostgreSQL to ClickHouse POC** for AssortSmart planning analytics, benchmarking the **23.7M row** Order Batching metric at **3.86s on ClickHouse vs 3m 40s+ on PostgreSQL (60x)** and bulk loads at **5.9M vs 250K rows/sec**.
- Designed the ClickHouse read path for Order Batching, keeping writes on PostgreSQL, syncing hot fact tables in near real time, and routing post save reads through **Redis (30s TTL)** so planners see their own saves instantly.
- Cut 10K row updates on a **29M row fact table from 39s to 7s** with partition scoped merges and **ReplacingMergeTree** versioned updates, and compressed a 17M row fact table **7.8x** to 433 MiB.
- Tech Used:** Python, FastAPI, LangGraph, MCP, ClickHouse, PostgreSQL, Redis, Kafka, GCP, Docker.

Software Development Engineer 2 UBER (via EPAM Systems), Bangalore July 2024 – May 2026

Financial Risk Management (FRM) Scoping Platform (Uber Finance)

- Built Uber's **FRM Risk Scoping platform (FastAPI, MySQL, React)** across **8 screens and 30+ REST APIs**, targeting a **70% cut** in manual scoping work at **\$340M group materiality**, with output feeding PwC audit work papers.
- Designed an **11 table SQLAlchemy 2.0 MySQL schema** behind layered handler, service, and repository code with header based auth.
- Migrated Sheets backed reconciliation to **MySQL v2 APIs** validating HFM against public **10-Q** data, fixing a latent column aliasing bug.
- Led 3 engineers** and grew the Bazel tested service to **1,100+ pytest unit tests** gating CI.
- Tech Used:** Python, FastAPI, SQLAlchemy 2.0, Pydantic, MySQL, Bazel, pytest, Docker.

Menu Ingestion & Automation Platform (Uber Eats)

- Built a **Kafka, Flink, and Spark** pipeline for **30K+ menus/month**, cutting onboarding from 24h to 2h and saving **\$600K+ annually**.
- Built a **RAG and Gemini** extraction pipeline for PDF and image menus, reaching **98% fidelity and 100% schema consistency** in offline evaluation.
- Raised ingestion success from about 60% to **95%+** via IP rotation and proxy pools, and built **Pinot** dashboards cutting detection from hours to minutes.
- Tech Used:** Python, Apache Kafka, Apache Flink, Apache Spark, Apache Pinot, Selenium, Gemini, GCP, Docker.

Software Development Engineer 2 MASTERS INDIA, Noida December 2022 – June 2024

GST Compliance & E-Invoicing Platform

- Led a **PHP to FastAPI microservices** migration and **mentored 2 engineers**, cutting p95 latency from **1.2s to 300ms** for 1,500+ enterprise clients.
- Scaled bulk imports to **100K+ transactions and 1M+ daily**, raising sustained throughput from **700 to 4,000 requests/min**.
- Added **Redis caching** cutting redundant database reads 30%, and shipped audit logs that reduced client churn **15%**.
- Set up **ELK and New Relic alerting** cutting incident triage 70%, and raised test coverage from **35% to 82%** at a 98% deployment success rate.
- Tech Used:** Python, FastAPI, PostgreSQL, MongoDB, Redis, Elasticsearch, Celery, Docker, ELK, New Relic, AWS.

Software Development Engineer GEEKSFORGEES, Noida August 2021 – November 2022

- Migrated PHP services to **Django (Python)**, improving reliability for **10K+ daily queries**.
- Built voting and pinning APIs on MySQL, MongoDB, Redis, and Elasticsearch, lifting **premium subscriptions 15 to 20%**.
- Built influencer analytics and cron pipelines for video processing, raising **course sales 30%** and operational efficiency 70%.

EDUCATION

B.Tech (Information Technology), CGPA 7.7/10 Institute of Engineering & Technology, Lucknow July 2017 – June 2021

ACHIEVEMENTS & CERTIFICATIONS

- Ranked **2260 / 37,000+** in the Google Code Jam Qualifier Round 2021.
- Finalist (top 3 nationally) at the **Smart India Hackathon 2020 (MHRD)** and participant in the **Global AI Hackathon** at EPAM.
- Certifications:** **HackerRank Problem Solving**, **LangChain Academy**.